

Report

Caption-Quality and Shot-Boundary Evaluation of a Video-Curation Pipeline

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1 Overview: what is measured and why

This report documents the first end-to-end run of a new evaluation harness for a video captioning and segmentation pipeline (NVIDIA `cosmos-curate`, adapted for robot- and human-activity video). The pipeline splits long videos into windows, generates a caption per window with a vision-language model (VLM), and flags suspect captions with VLM judges. The harness scores three independent questions on that output:

1. **Grounding** — does a caption describe the action actually happening in its window? Measured both lexically (exact word overlap) and semantically (sentence-embedding match), because a caption can be correct in meaning while using different words.
2. **Judge reliability** — how consistently do the two VLM judges flag wrong captions, and do they agree?
3. **Shot boundary** — how well do the segmenters recover the ground-truth (GT) action boundaries?

A recurring design principle is that every metric is built to be *not gameable*: scores should not rise simply because captions are longer, nor be silently tied to a single embedding model. To prove the harness measures real difficulty rather than artefacts, each primary experiment on the main dataset is paired with a *cross-domain control* on a second dataset (Sections ?? and ??).

2 Datasets and models, and why they were chosen

AgiBotWorld-Alpha (primary). 2,587 robot-manipulation videos, 15,526 caption windows, 14,996 GT action segments. This is the pipeline’s primary target: a tabletop robot arm picking and placing objects. The distinguishing signal is *object and action identity*, which is exactly what captioning is expected to capture, so it is the natural stress test for grounding.

InHARD Online (shot-boundary control). 38 full human industrial-assembly sessions (~400–650 s each, continuous fixed-camera footage — *not* pre-cut clips), with 4,803 GT action segments at real wall-clock timestamps. It was chosen because it is genuine, non-robot footage with *real visual variation* (a worker reaching to different bins, handling visibly different parts), already available on disk with timestamped GT. It is the counterfactual to AgiBot: if detectors fail on AgiBot but work here, the AgiBot failure is a domain property, not a harness bug.

YouCook2 (caption-grounding control). 4,471 cooking-video caption windows with GT recipe-step text. Cooking is a domain web-trained models know well, so a correct harness should report *higher* grounding here than on robot video. It tests whether the caption metrics track domain difficulty rather than being saturated or broken.

Caption models. The model under test is **Qwen-2.5-VL-7B**. A stronger model, **Qwen3-VL-30B (FP8)**, was added as a comparison point (Section ??). Judges are Gemma4-E4B and VCInspector-7B (VCI-7B); semantic grounding uses two text encoders, MiniLM (small) and BGE-large (strong).

3 Shot-boundary detection

Goal. Split each long video into segments aligned with GT action boundaries. **boundary-F1@0.5s** counts a predicted boundary correct if within 0.5s of a GT boundary; **segment-F1@IoU** measures segment overlap; **overSeg** is $\text{predicted} \div \text{GT segment count}$.

Table 1: AgiBotWorld — segmenters vs. GT action boundaries.

detector	nPred	bF1@0.5s	bP@0.5s	bR@0.5s	segF1@0.1	segF1@0.5	overSeg
fixed-stride (baseline)	2,587	0.039	1.000	0.039	0.354	0.056	0.237
TransNetV2	3,237	0.020	0.778	0.043	0.369	0.065	0.289
semantic SigLIP2	3,235	0.020	0.779	0.043	0.369	0.065	0.289

Result on AgiBot (the key negative finding). TransNetV2 and SigLIP2 both score boundary-F1 below 0.04 — *no better than naively cutting at fixed intervals*. This is not a bug. TransNetV2 and SigLIP2 detect *visual* scene changes; AgiBot action boundaries are *semantic* — the arm keeps the same scene and background while switching from “pick” to “place”, so there is no visual cut to detect. TransNetV2’s boundary-precision is high (0.78) but recall is only 0.04: when it fires near a GT boundary it is usually right, but it misses $\sim 96\%$ of them. The honest implication is that GT-aligned or fixed-stride windows are the correct choice for this domain; visual shot detection adds nothing, and the harness now proves this quantitatively.

Cross-domain control on InHARD. To confirm the failure is a domain property, the same detectors and harness were run on InHARD. AgiBot rows are repeated for side-by-side reading.

Table 2: Cross-domain control: same detectors, robot vs. human-assembly footage.

detector	dataset	nPred	bF1@0.5s	bP@0.5s	bR@0.5s	segF1@0.5
TransNetV2	AgiBot (robot)	3,237	0.020	0.778	0.043	0.065
TransNetV2	InHARD (assembly)	292	0.047	0.410	0.025	0.001
SigLIP2	AgiBot (robot)	3,235	0.020	0.779	0.043	0.065
CLIP-L	InHARD (assembly)	1,684	0.236	0.417	0.168	0.146
V-JEPA2-L	InHARD (assembly)	1,651	0.221	0.392	0.158	0.131
SigLIP2	InHARD (assembly)	1,533	0.220	0.420	0.154	0.134
DINOv2-L	InHARD (assembly)	1,386	0.201	0.409	0.136	0.109

Hypothesis confirmed. The semantic encoders jump roughly $10\text{--}12\times$ in boundary-F1 when moved from AgiBot to InHARD (SigLIP2 $0.020 \rightarrow 0.220$; CLIP-L reaches 0.236, and 0.364 at the looser 1.0s tolerance). Same code, same detectors — only the domain changed. This proves the AgiBot result is intrinsic to that domain: when boundaries are *visually expressed*, the encoders recover them; when they are not, nothing fires. Two finer points: (i) the pure pixel-cut detector (TransNetV2) barely moves even here ($0.020 \rightarrow 0.047$), because continuous single-camera assembly has almost no hard cuts — it is the *semantic* encoders, not the cut detector, that capture gradual appearance shifts; (ii) absolute recall stays modest (~ 0.16) because InHARD is extremely fine-grained (~ 126 actions/video, ~ 4 s each), so precision (~ 0.42) and the relative ordering ($\text{CLIP} \gtrsim \text{V-JEPA2} \approx \text{SigLIP2} > \text{DINOv2} \gg \text{TransNetV2}$) are the reliable signal.

4 Caption grounding: lexical vs. semantic

Does the caption mention the action the video actually shows? Lexical scoring uses exact content-word overlap with the GT action text (strict, penalises synonyms). Semantic scoring splits the caption into clauses and matches each GT action to its best clause by embedding cosine; recall asks “is each GT action covered?” and precision asks “is each clause the model wrote actually supported?” (penalising padding).

Table 3: AgiBotWorld — caption grounding (Qwen-2.5-VL-7B).

metric	value	what it measures
Lexical action hit-rate	0.086	fraction of actions with recall ≥ 0.5
Lexical action-F1	0.107	headline lexical
Semantic recall (MiniLM)	0.411	clause-level meaning match (small encoder)
Semantic F1 (MiniLM)	0.400	length-robust headline (small)
Semantic recall (BGE)	0.616	clause-level meaning match (large encoder)
Semantic F1 (BGE)	0.610	length-robust headline (large)
Granularity-aware recall	0.198	bidirectional containment match

The lexical-semantic gap is the story. Lexical action-F1 is only 0.107 but BGE semantic-F1 is 0.610: Qwen describes the right action (“grasps the object”, “places it in the bin”) but rarely reuses AgiBot’s exact verbs and nouns. A purely lexical metric would badly under-rate the captions. The metric is robust by construction — the correlation between semantic-F1 and caption length is -0.069 (essentially zero), so the score cannot be inflated by writing more words — and the two encoders are reported side by side (BGE 0.610 > MiniLM 0.400) so the result is not silently tied to one embedding model.

Cross-domain control on YouCook2. The same caption harness was run on cooking video, where every metric is expected to improve if the harness tracks real difficulty.

Table 4: Cross-domain control: caption harness on robot vs. cooking video.

metric	AgiBot (robot)	YouCook2 (cooking)	direction
quality_score	0.583	0.676	higher (easier domain)
Semantic F1 (BGE)	0.610	0.665	better grounding
Semantic F1 (MiniLM)	0.400	0.407	$\approx$
Lexical action-F1	0.107	0.246	$2.3\times$
Unique caption ratio	0.713	0.945	less repetition
Gemma4 / VCI-7B flag rate	0.563 / 0.772	0.469 / 0.360	fewer flagged wrong
Judge Cohen’s κ	0.108	0.148	judges agree more

Result confirmed. Every metric moves in the expected direction: on cooking the captions are more diverse, better grounded both lexically and semantically, flagged less often, and the composite **quality_score** rises $0.583 \rightarrow 0.676$. The harness is neither saturated nor flat — it tracks domain difficulty. One honest caveat: the lexical gap stays large in *both* domains (YouCook2 lexical-F1 0.246 vs. semantic-F1 0.665), confirming that lexical overlap under-rates captions everywhere and the semantic metric is the right headline.

5 Judges, consistency, and idle handling

Two VLM judges watch each window’s video and independently flag captions they think are wrong (no GT is shown to them). A separate consistency block needs no GT at all: it checks whether neighbouring windows of the same clip agree.

Table 5: Judge agreement and reference-free consistency (AgiBotWorld).

metric	value	meaning
Gemma4-E4B flag rate	0.563	fraction of captions called wrong
VCI-7B flag rate	0.772	fraction of captions called wrong
Raw agreement	0.585	fraction both judges agree on
Cohen’s κ	0.108	agreement beyond chance
Both-incorrect rate	0.460	high-confidence error captions
Object continuity	0.273	adjacent windows share the object
Action continuity	0.200	adjacent windows share the verb
Contradiction rate	0.053	clip captions disagree (lower better)
Idle-caption recall	0.785	says “nothing happening” when idle
False-idle on active	0.040	wrongly idle on active (lower better)

The judges disagree a lot, and surfacing this honestly is itself a result. VCI-7B flags 77% of captions, Gemma4 only 56%, and $\kappa = 0.108$ means agreement is barely above chance; they catch different failure modes (VCI aggressive, Gemma4 conservative). Using both yields a high-confidence “both-incorrect” set (46% of windows) far more trustworthy than either alone. Idle handling works well: genuinely idle windows are labelled idle 78% of the time and active windows are wrongly called idle only 4%, so idle windows do not pollute the action metrics.

6 Text–video alignment (reference-free)

Unlike grounding (caption vs. GT *text*), this scores each caption directly against the *video frames* with SigLIP2 — no GT needed — and is the only grounding signal usable when no GT action text exists. All 15,526 windows were scored.

Table 6: Text–video alignment, AgiBotWorld (SigLIP2).

metric	value	meaning
caption↔video sim (mean)	0.216	match to frames (std 0.033)
GT-text↔video sim (mean)	0.137	ceiling: how well the correct label grounds
caption – GT	+0.080	caption aligns better than the GT label
retrieval median rank	387 / 15,526	discriminativeness (random $\approx 7,763$)
R@1 / R@10	0.004 / 0.034	rarely finds its own clip exactly
within-clip norm. rank	0.446	$\approx$ random (0.5) inside a clip

Two observations both *confirm* a known limitation rather than contradicting it. First, the caption scores *higher* against the frames than the GT label does (+0.080): SigLIP2 is web-trained, so a vivid descriptive caption matches the general scene better than a terse GT label (“grasp”), even when the specific object may be wrong. Second, retrieval is far above random *globally* (median rank 387 vs. 7,763) but essentially random *within a clip* ($0.446 \approx 0.5$) — the encoder can tell a fridge clip from a supermarket clip but cannot tell two adjacent windows of the same clip apart. This is the familiar “works at task level, fails at fine-grained object level” pattern; text–video is therefore kept as a coarse secondary check, with the VLM judges and text–text semantic metrics remaining primary.

7 Caption-model comparison: Qwen3-VL-30B (FP8)

A stronger model, Qwen3-VL-30B-A3B-Instruct (FP8), was added to extend the caption comparison. It initially *failed* on the A100 GPU: the Ampere architecture has no native FP8 support, so the inference engine fell back to a kernel that dequantises the FP8 weights to BF16 at runtime (~ 60 GB), exhausting the

40 GB GPU and leaving 0 GB for the key-value cache, so the engine could not start. The fix was to run on an **H100** (Hopper has native FP8): the weights load at their true ~ 31 GB, leaving ample room for the cache, and a single GPU suffices. The run then completed in ~ 7 minutes, captioning all 30 test videos cleanly with high-quality, structured output (e.g. “*The robotic arm moves forward over a display of toy vegetables, then lowers its gripper to grasp a purple eggplant.*” with explicit action/object fields). This unblocks ranking Qwen3-VL-30B against Qwen-2.5-VL-7B on the grounding and judge metrics above.

8 Efficiency and caption statistics

Table 7: Caption statistics and throughput (AgiBotWorld, Qwen-2.5-VL-7B).

statistic	value	note
Caption windows	15,526	one caption per window
Unique caption ratio	0.713	lower = more repetitive
Mean / median words	17.6 / 18	caption length
Vocabulary size	923	distinct words used
Most-duplicated caption	$\times 77$	repeated verbatim
Caption compute / caption	0.309 s	true per-stage compute
Est. caption tokens / s	74.2	throughput proxy

9 Verdict

The harness behaves exactly as designed and its numbers are internally consistent. Lexical scores are far below semantic scores (synonym tolerance working); the semantic score is length-robust (not gameable); idle windows are cleanly separated; the judges’ low agreement is surfaced honestly; and shot detection is shown to be no better than fixed-stride on a semantic-boundary domain. Crucially, both cross-domain controls succeed: detectors recover $10\text{--}12\times$ more boundaries on InHARD’s visually varying footage, and the caption metrics improve across the board on YouCook2’s easier cooking domain — proving the harness measures genuine difficulty rather than artefacts. For the captions themselves, Qwen-2.5-VL is semantically decent (BGE semantic-F1 0.61) and self-consistent (5% contradiction) but lexically loose and somewhat repetitive (unique ratio 0.71): it gets the action right and the exact object wording less so. The harness produces one comparable `quality_score` per model and is ready to rank stronger captioners such as Qwen3-VL-30B.